

Figure 1: Simple app design principle

same *main.xml* in this folder, so that when the device orientation changes, the application layout changes from portrait to landscape, and vice-versa.

Now that we have the layout defined, what remains is the code behind it. Let's start with the *FragmentTestActivity* class. In the *onCreate()* method, set the list adapter to hold a string array of EFY Group magazine titles. Also set the listener on the list items, so that you can perform some action when an item from the list is clicked.

```
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.main);
    ListView l = (ListView) findViewById(R.id.number_list);
    ArrayAdapter<String> magzTitles = new ArrayAdapter<String>(getApplicationContext(),
        android.R.layout.simple_list_item_1,
        new String[]{"Electronics For You",
            "Linux For You",
            "Facts For You"});
    l.setAdapter(magzTitles);
    l.setOnItemClickListener(this);
}
```

In the *onItemClickListener()* method, perform the main task of managing the *fragment*. Obtain the instance of the *fragment* by passing the position of the clicked item. Now, you need to replace the *fragment* element in *main.xml* with the new *fragment TestFragment*, which has a meaningful UI associated with it. To

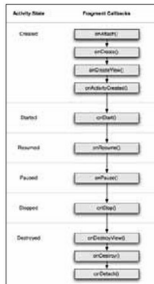


Figure 2: Fragment callbacks during its lifecycle



Figure 3: Package structure



Figure 4: The main.xml file

accomplish this, get the instance of the *FragmentManager* class. This API allows you to add, remove and replace a fragment programmatically. Replace the *R.id.the_frag*, which corresponds to the *<fragment>* element of *main.xml*, with the newly created fragment *'f'*. The *setTransition()* method signifies the kind of transition that happens with the fragment. Some of the transition values are shown in Table 1.

Table 1: Some transition values and explanations

TRANSIT_FRAGMENT_CLOSE	Fragment is being removed from the stack.
TRANSIT_FRAGMENT_FADE	Fragment should simply fade in or out; i.e., there should be no strong navigation associated with it, except that it appears or disappears for some reason.
TRANSIT_FRAGMENT_OPEN	Fragment is being added onto the stack.
TRANSIT_NONE	No animation for transition.
TRANSIT_UNSET	Not set-up for a transition.

The *addToBackStack()* method adds the fragment transaction to the fragment stack, so that when the back button is pressed on the device, you go to the last transaction of the fragment, and do not exit the application. After the transaction is set up, commit it, as follows:

```
public void onItemClick(AdapterView<?> parent, View view, int position,
    long id) {
    Fragment testFragment = new TestFragment(position+1);
    FragmentTransaction ft = getSupportFragmentManager().
```